

Syllabus for MATH 202X: Future-Proving for Math Majors

Instructor: Alp Uzman, Ph.D.

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1 Staff and Meeting Hours

Instructor: Alp Uzman, Ph.D. (uzmancoaching@gmail.com)

Lectures: Three hours per week, for 15 weeks in total. More details to be determined, including whether or not the lectures will be synchronous or asynchronous, in-person or online. Either way, edited and annotated recordings of lectures will be regularly uploaded to the following Youtube playlist:

<https://www.youtube.com/playlist?list=PLaZPOp1Bmku8>.

Course Webpage: As the course webpage we'll use the following Github repo:

<https://github.com/AlpUzman/FutureProving>.

Office Hours: By appointment

2 Course Description & Prerequisites

Future-Proofing versus Future-Proving Roughly speaking, **future-proofing** is a phrase that broadly signifies the collection of anticipatory practices to make things more resistant to the effects of the passage of time. The phrase "future-proving" is a twist on this; while certainly future-proving

is in the same **family of resemblances** as future-proofing, it is not quite the same. This course is not about making mathematics "future-proof" or even "future-resistant" as in a "water-proof coat" or a "**water-resistant watch**". Instead, the course is roughly about the exact opposite.

We Haven't Seen Anything Yet The starting point of this course is the ramifications for human mathematics of the recent developments in AI, such as the advances in **formalization**, **counterexample generation** or **problem solution**. It seems the value of proving things complicated enough that in my eyes provides sufficient justification to call myself a mathematician the old-fashioned way (with no ultimately removable help from a computer), like I trained in as someone who was in college from 2011 to 2016 and in graduate school from 2016 to 2023 is fundamentally changing. When I was in grad school, I did not have internet at my house, because I did not need it to work (**and then the funniest thing happened**, and such behavior would not do anymore).

When I was in college as a math major, I would tell my engineering friends that I don't compute the types of nasty integrals that they were forced to take during their classes (I also had no interest in being a technical person apart from a "200 proof" (aka "pure") mathematician at the time, which presumably is a juvenile perspective I changed since then). Then as I was progressing in my studies, I realized that my understanding of a calculation started expanding, possibly based on the realization that in fields that use mathematics (say), often they would say they are "proving a theorem", whereas for me the very same thing was nothing but "a calculation (by hand) accompanying an observation". With the latest developments downstream from AI, the line gets even more blurry.

Fast-forwarding time to say around the time the Riemann hypothesis is cyberproven (with an accompanying Lean verification as well as a collection of company-and-institution-backed NFT's and the cryptocurrency RIEMANNHYPE), presumably "**I don't prove things: I'm a mathematician!**"

would be a fairly common exclam, as it is nowadays with programmers who don't code. (You can take this as a joke, cruel and bitter or otherwise: I remember joking about subsidized theorems while in grad school, inspired by DFW's idea of "subsidized years"; we see now theorems attributed, rightfully or wrongfully, to companies.)

Right around the same time, an anonymous account on Youtube posts an ASMR video of a complete proof of Fermat's Last Theorem, automatically rendered using **Pink Trombone**. The compute implied and the sheer length of the video barely manages to make news.

This is a world of mathematics that not only the average graduate student in the year 2025 but also human mathematicians said to be the Mozart, Aristotle, Goethe, Attila and Thanos of mathematics combined are overwhelmed as a regular state of affairs and hence as a rule start any speech by an appropriate disclaimer, together with an acknowledgement of ample AI use with an estimate on the total cost of tokens used to prepare said speech, to exemplify good practice, possibly among other things.

What is at Stake? Insofar as we have indeed not seen anything yet, this course is an invitation for the younger generations to start proving the future.

For a fairly long time I have been confident in the value of mathematics as an embodied human activity. Following through, over the past few months, I found myself thinking that it is a human right to learn math from human mathematicians. On the other hand following through the politico-economical logic of academic institutions, I started realizing that the experience of learning and doing mathematics as I and previous generations have done is going to survive only as a luxury (or anomaly); similar to how nowadays it is a luxury to use fountain pens.

Accordingly, I have been trying to come up with ways out of this situation, in my own humble ways (I also have a bachelors degree in philosophy, and while dire, this problem actually interests me as a problem to be solved

separately, as opposed to many problems in mathematics). Eventually I realized how prone I was in thinking about myself in the past tense, and how much my own outlook is colored by my experiences ("maybe it's not necessarily a human right to learn math from humans?"), which then led to the realization that in fact we as human mathematicians and educators should be active in providing the younger generations the opportunity to start asking what math looks like at the end of history.

This course is my attempt at providing one such opportunity. The overall design of the course reflects my biases, and there are many other kinds of courses that focus on mathematical future-proving. It just turns out that if doing math does not look like what it looked like when I was a student, it might possibly look like playing a video game or reading a Calvino story to me.

Prerequisites Not much in this course actually will involve mathematics in the old fashioned way, and the only official prerequisite is "Instructor Consent". While there is some flexibility, to avoid overdetritorialization I will likely ask for evidence that you have mastery over the standard calculus series as well as the introductory linear algebra course (or instead the introductory differential equations course). In addition, I will ask for evidence that you have taken at least one non-STEM course, at least two upper-division STEM courses, and at least one proof-based math course where you were expected to produce some proofs without typing. In case there are collisions between these, it is likely I will give consent for you to take this course.

Doubly-Ergodic Gambit I also intend to use this syllabus as part of my academia applications in the 2026-2027 academic year, as I don't think I have anything else to offer that can give me the competitive advantage (this is the other side of the medallion that has this syllabus on this side). In the past I have had the opportunity to design an eccentric math course

("Introduction to Mathematical Research") before, and if I do get hired and have the opportunity to teach this course, it is likely the actual syllabus will be somewhere between this and the syllabus of the course I designed before.

This certainly is a gambit, and hence for now it's an instance of "course-fiction" (inspired by Borges' reviews of unwritten works). Insofar as this is fiction, I ought to ask for job security and reimbursement of moving costs also. Paradoxically, the previous official narrative goes that the opportunity cost implied by me being able to design this syllabus is apparently high enough for me to not be hired as a non-precarious mathematician at a professional institution. Note that writing this syllabus took me at least about 15 years and it's unlikely I'll be able to come up with something comparably innovative, even with ample amount of tokens, so buyers beware.

There is also a separate but related intrinsic gambit involved: even if the previous gambit backfires, my hope is that this syllabus will come off as sincere enough to get some people thinking in a humanist way, despite the implied conflicts of interests and hyperstitious effects.

Variants There are possible various variants of this course based on the shift in emphasis:

MATH 202X. π : Phenomenology for Math Majors

MATH 202X. φ : Fiction for Math Majors

MATH 202X. γ : Game Making for Math Majors

MATH 202X. ϵ : Economy for Math Majors

MATH 202X. η : History for Math Majors

MATH 202X. v : Animation and Film Making for Math Majors

Note that this course has the cross-code MATH 202X. σ (σ for "Self").

3 Expectations

As a student, you are expected to:

- attend all lectures,
- complete weekly reports,
- complete a final project.

As a rough estimate, you should expect a workload of approximately 2 to 3 hours per lecture hour every week. Due to the open-ended nature of the course, your actual workload may vary depending on your level of engagement.

4 Learning Objectives

AU: TODO

5 Books & Other Resources

AU: TODO

6 Lectures

Attendance at all lectures is expected. Attendance at random lectures will be taken and will have an effect on the final grades. The number of lectures when attendance will be taken will also be random.

Lecture Recordings Video recordings of lectures, as well as supplementary material in video form, will be uploaded to the following Youtube playlist:

<https://www.youtube.com/playlist?list=PLaZPOp1Bmku8>.

7 Tentative Schedule

AU: TODO

The course's subject matter is divided by weeks within a 15-week academic calendar as follows.

Week 1: Atemporality and Hauntology for the Passing Mathematician https://librarynth.net/atemporality_for_the_creative_artist <https://web.archive.org/web/20100211180606https://transmediale.de/en/keynote-bruce-sterling-us-atem>

<https://transmediale.de/en/keynote-bruce-sterling-us-atem>

Week 2: Lean Games Natural Number Game. Real Analysis Game. Lean Game Game. Dependent Type Theory.

Week 3: Mathematicians Doing Non-Mathematics, Mathematicians Non-Doing Mathematics Rota. Manin. Weyl. MacLane. Poincaré. Gromov. Marcolli. Grothendieck. Halmos. Krantz. Tao. Sokal.

Week 4: Introduction to Phenomenology Definition(s) of phenomenology. Some Famous Phenomenologists.

Week 5: Static Phenomenology

Week 6: Dynamical Phenomenology

Week 7: Manifold Hypothesis Fefferman and company: Reconstruction.

Week 8: Cyberfiction Neverending Story. Markson. Borges. Calvino. Ballard. Sterling. Paranoia Agent. Evangelion. Serial Experiments Lain.

Week 9: Cybertext and Ergodic Literature Ergodic Ergodicity.

Week 10: Ludology and the Trickster Archetype Huizenga. Carse's Infinite Games. Fink.

Week 11: Esolangs and Livecoding Strudel. Delay Lama. Pink Trombone.

Week 12: Accelerationism versus Accelerationist Realism

Week 13: NFT's, Blockchain and Cryptocurrencies

Week 14: "Compute" as a Noun: All that is Solid Melts into Tokens

Week 15: Post-Manifold Hypothesis Adams-Krantz. PageRank, EdgeRank etc.. Freedman and company: Compression. Barnsley and company: Fractal Compression.

8 Talks

The contents of the lectures reflect my biases. To further deterritorialize, there will be various talks from experts in some other possibly related fields (for instance 3D printing as well as simulations) that are not as represented in the schedule. The speakers, talk titles and abstract and time and space will be released in time. The students will have the opportunity to nominate speakers.

9 Weekly Reports

Students are required to submit a weekly video report each week, totaling 15 weekly reports. Each weekly report is due by 11:59 PM on the Sunday ending the week. Late submissions will not be accepted unless you contact me with a valid excuse before the deadline. A certain number of weekly reports with lowest scores may be dropped at the end of the semester.

Content The content of a weekly report will depend on the student's personal preferences, as long as it is within the scope of the class and within reasonable limits. The length of a weekly report should be between 5 to 10 minutes, and your face must be visible in every single frame. A weekly report may include:

- critique of a structural element of this course
- a response to a topic or work that was discussed in lectures
- an advancement in mathematics, downstream from AI or not
- a rhetorical analysis of a mathematical discussion
- experiments toward finding your final project
- your progress toward your final project

Weekly projects will be collective: Collaboration with other students is allowed and encouraged. However, in case of content overlap in the reports of two different students, the earlier submission will be assessed, and the later submission will receive an automatic zero for the overlapping content.

Technical Specifications All weekly reports must be 5 to 10 minute long videos in MP4 format, at a resolution of at most 1080p and framerate of at

most 24 fps. A convenient and free video editing program is **Davinci Resolve**. As familiarity with video editing and Davinci Resolve is not assumed, Weekly Report 0 is reserved for practice.

Grading Criteria Weekly reports will be graded on accuracy, validity and presentation/documentation.

Submission Details Detailed submission instructions will be provided later as specification documents.

10 Final Report and Presentation

AU: TODO

Instead of a final exam, there will be a final report and an in-class oral presentation based on the final report. Here are the relevant deadlines:

- **Declaration of topic and primary source(s):** by the Sunday of Week 9
- **First draft for final project:** by the Sunday of Week 12
- **First draft and plan for presentation:** by 11:59 PM on November 24 (Sunday of Week 13)
- **Presentation:** in class on December 3 or December 5 (the two classes in Week 15)
- **Final report:** by 11:59 PM on December 8 (Sunday of Week 15)

Broadly speaking, the topic should be according to the scope of the class. The complexity and depth of the project depends on personal preference (within reasonable limits) and could vary from an exposition or expansion of a certain chapter in one of the books to an exposition of a research article,

to generalization of a simpler problem, and even discussion or solution of an original problem. The final report must be prepared using Latex and should be between 5 to 10 pages, not counting the acknowledgements and references. The final report and presentation will be collective, in the sense that while collaboration with other students is allowed and encouraged, only one student will be assessed per report.

The presentation may be a **chalk talk** or digital as a **Beamer** presentation. The length of the presentation will be such that the total time for the two lectures in Week 15 is equidistributed among students.

11 Grades

The final letter grades will be determined according to the following weights and cutoffs. The cutoffs below are not definitive; based on the overall performance they may be adjusted at the end of the semester.

Assignment	Weight
Attendance	25%
Weekly Reports	35%
Final Project	40%
Total	100%

Grade	Percent
A	90%
A-	85%
B+	80%
B	75%
B-	70%
C+	65%
C	60%
C-	55%
D+	50%
D	45%
D-	40%
E	else

12 Exams

There will be no midterm exams nor a final exam.

13 Accessibility

AU: TODO

The university seeks to provide equal access to its programs, services, and activities for people with disabilities. If you will need accommodations in this class, reasonable prior notice needs to be given to the appropriate office. The office will work with you and the staff to make arrangements for accommodations.

14 Inclusivity and Safety

AU: TODO

Title IX makes it clear that violence and harassment based on sex and gender (which includes sexual orientation and gender identity/expression) is a civil rights offense subject to the same kinds of accountability and the same kinds of support applied to offenses against other protected categories such as race, national origin, color, religion, age, status as a person with a disability, veteran's status or genetic information.

If you or someone you know has been harassed or assaulted, you are encouraged to report it to the appropriate authorities.

15 Academic Honesty

AU: TODO

You are expected to adhere to the university policies regarding academic honesty. This means that ultimately all work you submit must be your own, created without unauthorized assistance, and all external resources, including generative tools and computer algebra systems, must be properly cited and documented within your submissions. Any student who engages in academic dishonesty or who violates the professional and ethical standards may be subject to academic sanctions as per the university's student code.

Generative AI, Computer Algebra Systems In this course, you may use generative tools like ChatGPT as well as computer algebra systems such as MATLAB under specific guidelines. These tools are permitted to guide your understanding of topics. If applicable, you must acknowledge your use of such tools in your submissions. You are responsible for ensuring the validity, accuracy, and relevance of your submissions, whether or not you use these tools. While encouraged, the use of such technology is not required for success in this class.

Regret Clause If you commit an unreasonable act but bring it to the staff's attention within 48 hours of the relevant submission, sanctions may be limited to that submission only, rather than leading to further disciplinary action. This clause will not be applied in the case of repeated violations.

16 Acknowledgements

While I am not against the use of AI, and in the past I have used AI for the purposes of retouching syllabuses I designed¹, I have not explicitly used AI to prepare this syllabus(-fiction), both as a **design choice** and so as to not give in to **accelerationist realism**. I have benefited from conversations with and encouragement from L.M.W. and P.T., thank you both.

¹Including the **course** whose syllabus I recycled to prepare this syllabus.